QSG & DIG Innovation Incubator

Bank of America Merrill Lynch

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CMU Data Science Certification Program

As was discussed in the November Town Hall and the April Newsletter, the Innovation Incubator has been working to setup a range of learning opportunities and resources, catering towards a variety of individual interests and learning habits. Today, we are very excited to announce a rewarding and demanding academic program created directly as a collaboration between Leif and Carnegie Mellon University – **CMU Data Science Certification Program.**

The Data Science Certificate Program at CMU will be available exclusively to BofA in 2020-2021 and consists of six courses in the Master of Science in Computational Finance (MSCF) program:

1. Probability Prep
2. Financial Data Science I
3. Financial Data Science II
4. Machine Learning I
5. Machine Learning II
6. Financial Time Series Analysis

See the Appendix for additional details on the courses. Proper academic CMU credits at the Master's level will be earned upon completion of these courses; a Data Science Certificate will be granted upon completion of all required courses, provided that a 3.0 average (or better) is maintained. Exams are either in-person or proctored at a remote location. Total duration of the program is 4 mini-semesters, spanning Fall 2020 through Spring 2021; see the table on Page 2 for the 2020-2021 academic calendar.

The lectures are given during business hours, they are recorded, and the recordings are available to enrolled students. Students may also interactively follow lectures in real time from remote locations. The courses have websites where written discussions among the students, instructors and teaching assistants occur.

The admissions deadline is June 1, 2020. Approval from a Senior Quantitative Finance Manager (i.e., Leif's direct reports) is required before an application can be sent to CMU.

Once approved to submit an application, all prospective students must apply through this link: **https://apply.mscf.cmu.edu/portal/certificate**. Notice that the application will require a recommendation from your direct manager or above. As you consider application, please keep in mind that the program will require a considerable commitment from students.

The total cost is $13,200, of which 60% is paid in August 2020, and the remainder is paid in January 2021. Some fees will be paid directly by QSG, others will be paid by the student and later reimbursed in their entirety if the student maintains an acceptable level of academic achievement. The details of the reimbursement process vary by region and will be communicated separately. Notice, however, that some programs require applications for reimbursement to be submitted early in the study process (e.g. no later than 30 days after a course begins for the US program), so it will be important that students familiarize themselves early on with the reimbursement process.

Tentative Calendar, excluding Reimbursement Dates

Now - June 1	Consider commitment, discuss with manager, and obtain SQFM approval.
June 1	Deadline for on-line application to program.
July 1	Application decisions will be communicated by CMU.
August 3	Probability Prep class starts.
August 15	Tuition for fall classes due ($7,200).
August 31	Mini-semester 1 classes start.
October 26	Mini-semester 2 classes start.
October 28	Grades for mini-semester 1 are posted.
December 23	Grades for mini-semester 2 are posted.
2021	
January 11	Mini-semester 3 classes start. Second tuition installment ($4,800) due on or around this date.
March 10	Grades for mini-semester 3 are posted.
March 15	Mini-semester 4 classes start.
May 18	Grades are posted for mini-semester 4.

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This program is sponsored by the QSG & DIG Innovation Incubator whose mission is to foster idea generation and enable collaboration across the QSG & DIG teams globally.

Appendix: Course Descriptions

Probability Prep. The objective of this course is to introduce the basic ideas and methods of calculus-based probability theory and to provide a solid foundation for other MSCF courses based on probability theory. Topics include basic results on probability and conditional probability, random variables and their distributions, expected values, moment generating functions, multivariate distributions, transformations of random variables and vectors, laws of large numbers and the central limit theorem.

In 2020, the course will begin Monday, August 3, have two-hour lectures on Mondays, Tuesdays, Wednesday and Thursdays for three weeks, and conclude with a final exam the week of August 24. Probability Prep is a not-for-credit course and is free to students enrolled in the Data Science Certificate program. Although students do not receive credit for this course, in order to begin the program proper, they must either pass the course or pass an exemption examination.

Financial Data Science I. The first in a two-course sequence covering methods of extracting useful information from raw financial data. Focus is placed on tools of data exploration and mining, motivated by the handling of modern financial data sets. Topics include data cleaning, data visualization, linear regression, nonparametric smoothing techniques, and unsupervised learning. Methods are taught using the Python programming language.

This course is offered in the second half of the fall semester (mini-semester one).

Financial Data Science II. The second in a two-course sequence covering methods of extracting useful information from raw financial data. Focus is placed on fundamental tools of statistical inference useful for fitting models to financial data, including those with complex multivariate form. Topics include parameter estimation, uncertainty quantification, resampling methods, Bayesian inference, and discovery claims.

This course is offered in the second half of the fall semester (mini-semester two).

Machine Learning I. The first in a two-part sequence covering statistical machine learning aimed at quantitative finance. This first course covers tools and approaches for prediction, including both regression and classification. The focus is on understanding the foundations of the methods so that they can be both applied and modified. Topics include foundations of supervised learning, bias-variance tradeoffs, model validation and assessment, regularized and nonparametric regression, generalized additive models, support vector machines, and tree-based methods.

This course is offered in the second half of the fall semester (mini-semester two).

Machine Learning II. The second in a two-course sequence covering statistical machine learning aimed at quantitative finance. The course further covers methods for regression and classification, along with other advanced topics in statistics and machine learning. Topics will be drawn from boosting and ensemble methods, clustering, mixture models and topic modeling, natural language processing, Markov decision processes and reinforcement learning, and neural networks including the basic ideas underlying deep learning.

This course is offered in the first half of the spring semester (mini-semester three).

Financial Time Series Analysis. This course introduces time series methodology to the MSCF students. Emphasis will be placed on the data analytic aspects related to financial applications, with a view toward development of quantitative trading strategies. Topics studied in this course include univariate ARIMA modeling, forecasting, seasonality, model identification and diagnostics. In addition, GARCH and stochastic volatility modeling will be covered. At the end of the course, trading strategy development based on these models will be discussed. Reference texts (not required): Brockwell & Davis, Introduction to Time Series and Forecasting, 2nd edition, Springer (2002); N.H. Chan, Time Series: Applications to Finance, Wiley (2002).

This course is offered in the second half of the spring semester (mini-semester four).